

Full Paper Organization Guide

Monolingual Hate Speech Detection using DynamicFusionNet with Knowledge Distillation
Target: *Expert Systems With Applications* (Q1, Elsevier)

How this guide was built

This structure follows the organization conventions of recent ESWA papers (including the sample ESWA 2026 article you provided: Kovács et al., "Extracting available information from non-evaluable datasets..."), adapted to an applied deep learning paper. ESWA papers follow a consistent skeleton: **Abstract with keywords** → **Introduction ending with explicit contributions and a paper-organization paragraph** → **Literature review with an explicit research gap** → **Methodology** → **Experiments** → **Results with real-world validation** → **Summary/Conclusion** → **CRediT authorship statement** → **Declaration of competing interest** → **Data availability** → **References**. Your paper maps onto this skeleton as detailed below.

Key point about naming: Even though your files/notebooks say "bilingual", position the paper as **monolingual (English) hate speech detection** in the title, abstract, and framing. The MarianMT Bangla translation pipeline in the Streamlit app is presented only as a *deployment/accessibility feature*, not as a bilingual modeling claim — this keeps your claims defensible and consistent with your experiments.

0. Front Matter

0.1 Title

ESWA titles are descriptive and method-forward. Candidate patterns (pick one, refine wording):

- "DynamicFusionNet: An attention-based transformer ensemble with knowledge distillation for efficient hate speech detection"
- "Distilling a multi-transformer ensemble into a compact student model for accurate and deployable hate speech detection"
- "An explainable knowledge distillation framework for hate speech detection with cross-dataset generalization and real-time deployment"

Include your strongest selling points in the title: the ensemble/fusion architecture, knowledge distillation, and either explainability or deployment. Avoid vague words like "novel" in the title.

0.2 Abstract (150–250 words, single paragraph)

Write it in 7 sentences, in this exact order:

- **Context (1 sentence):** Hate speech proliferation on social media and the need for automated detection.
- **Problem/gap (1 sentence):** Large transformer ensembles achieve high accuracy but are too heavy for real-time deployment; compact models lose accuracy; explainability is rarely addressed.
- **Proposed method (2 sentences):** DynamicFusionNet — a teacher ensemble of BERT, XLM-RoBERTa, and DeBERTa fused via dynamic attention-weighted fusion, distilled into a BERT-base student via knowledge distillation (soft-label + feature-level losses, whichever you used).
- **Evaluation protocol (1 sentence):** Stratified 5-fold cross-validation, cross-dataset generalization on the Davidson dataset, SHAP-based explainability and misclassification analysis.
- **Headline results (1 sentence with numbers):** The distilled student surpasses the teacher (91.51% vs 90.16% accuracy) with roughly $N\times$ fewer parameters and $M\times$ lower latency — put the exact numbers.
- **Deployment (1 sentence):** The framework is deployed as a real-time Streamlit web application demonstrating practical applicability.

ESWA reviewers expect concrete numbers in the abstract. "Student surpasses teacher" is your hook — lead the results sentence with it.

0.3 Keywords (5–7)

Hate speech detection; Knowledge distillation; Transformer ensemble; Model compression; Explainable AI (SHAP); Cross-dataset generalization; Real-time deployment.

0.4 Highlights (Elsevier requirement, 3–5 bullets, max 85 characters each)

- A dynamic attention-fusion ensemble of three transformers is proposed as teacher.
- Knowledge distillation yields a student that outperforms its teacher (91.51%).
- Stratified 5-fold CV and cross-dataset tests on Davidson confirm robustness.
- SHAP analysis explains predictions and diagnoses misclassification patterns.
- A real-time Streamlit system demonstrates practical deployability.

1. Introduction (1.5–2.5 pages)

ESWA introductions follow a funnel: broad motivation → specific technical problem → limitations of existing solutions → your idea → explicit contribution list → paper roadmap. Write it as 6 paragraphs:

Paragraph 1 — Motivation:

Social-media scale, harms of online hate speech (psychological, societal, legal/regulatory pressure such as platform moderation requirements). One or two statistics with citations. Keep it to 4–6 sentences.

Paragraph 2 — Technical landscape:

Evolution from classical ML (TF-IDF + SVM/LR) → deep learning (CNN/LSTM) → transformers (BERT and variants) → ensembles. State that ensembles of large transformers currently give the best accuracy.

Paragraph 3 — The problem (your gap):

Three intertwined limitations: (i) ensemble teachers are computationally prohibitive for real-time moderation (parameters, latency, memory); (ii) models often fail to generalize across datasets/platforms (overfitting to annotation idiosyncrasies); (iii) black-box predictions hinder trust and moderation auditing. This paragraph justifies every component of your paper — KD answers (i), cross-dataset evaluation answers (ii), SHAP answers (iii).

Paragraph 4 — Your proposal:

Introduce DynamicFusionNet in 3–4 sentences: teacher = BERT + XLM-RoBERTa + DeBERTa with dynamic attention-weighted fusion; knowledge distillation transfers the ensemble's knowledge into a single BERT-base student; the student is validated with 5-fold CV, tested cross-dataset, explained with SHAP, and deployed via Streamlit. Mention the surprising finding (student > teacher) here — it is your strongest claim.

Paragraph 5 — Contributions (bulleted list, exactly as ESWA papers do):

- C1. A dynamic attention-based fusion ensemble (DynamicFusionNet) of three heterogeneous transformers for hate speech detection.
- C2. A knowledge distillation framework in which the compact student surpasses its teacher (91.51% vs 90.16%), analyzed and explained (regularization effect of soft labels).
- C3. A rigorous evaluation protocol: stratified 5-fold cross-validation with checkpoint/resume support and cross-dataset generalization on the Davidson dataset.
- C4. SHAP-based explainability and systematic misclassification analysis revealing which lexical cues drive predictions and errors.
- C5. A real-time, publicly deployable Streamlit system (with an auxiliary Bangla translation interface via MarianMT) demonstrating end-to-end practical applicability.

Paragraph 6 — Organization:

"The remainder of this paper is organized as follows. Section 2 reviews related work... Section 3 details the proposed methodology... Section 4 describes the experimental setup... Section 5 presents results and discussion... Section 6 describes the real-time deployment... Section 7 concludes the paper." (The ESWA sample paper you shared uses exactly this pattern at the end of its introduction.)

2. Related Work (1.5–2 pages)

Organize thematically with subsections, and end with an explicit research-gap paragraph — the sample ESWA paper does this by systematically reviewing prior approaches and then stating "This gap is addressed in this paper." Copy that rhetorical move.

2.1 Hate speech detection: classical and deep learning approaches

Feature-engineering era (n-grams, TF-IDF, lexicons) and early neural models (CNN, BiLSTM, attention). 1 paragraph.

2.2 Transformer-based and ensemble methods

BERT/Roberta/DeBERTa/XLM-R applied to hate/offensive/toxic detection; fusion and ensemble strategies (voting, stacking, weighted averaging, attention fusion). Position dynamic attention fusion against static fusion. 1–2 paragraphs.

2.3 Knowledge distillation and model compression for NLP

Hinton's KD, DistilBERT/TinyBERT/MobileBERT, ensemble-to-single distillation, and prior reports of students matching/exceeding teachers (born-again networks). This subsection sets up why student > teacher is plausible, so reviewers do not treat your key result with suspicion.

2.4 Cross-dataset generalization and explainability in hate speech detection

Known generalization failures across datasets (e.g., Davidson vs other corpora, annotation bias); LIME/SHAP/attention-based explanation studies for toxicity models.

2.5 Research gap and positioning

Close with a comparison table (Table 1): rows = 8–12 representative prior works, columns = {Ensemble teacher, KD, 5-fold CV, Cross-dataset eval, XAI, Real-time deployment}. Your row is the only one with all ticks. Then one paragraph: "To the best of our knowledge, no prior work combines... This paper fills this gap."

Table 1 (the gap table) is the single most persuasive object in Section 2 — reviewers scan it before reading prose.

3. Proposed Methodology (3–4 pages)

This is where you present each notebook's work as a formal component. Use math notation (define the classification task, loss functions with equations) — ESWA expects formal presentation even in applied papers.

3.1 Problem formulation

Define the task: given text x , predict label $y \in \{\text{hate, non-hate}\}$ (or your label set). Define notation for teacher T , student S , logits, temperature τ , etc.

3.2 Overall framework

Figure 1: end-to-end pipeline diagram — raw text → preprocessing → teacher ensemble training → distillation → student → evaluation (CV + cross-dataset + SHAP) → deployment. One paragraph walking through the figure.

3.3 Data preprocessing

From your notebooks: cleaning (URLs, mentions, emojis, normalization), tokenization per backbone, max sequence length, label mapping/remapping (document the remapped dataset scheme you used for cross-dataset alignment), train/val splits, class distribution handling.

3.4 Teacher: DynamicFusionNet ensemble

- 3.4.1 Backbone encoders — BERT, XLM-Roberta, DeBERTa: brief description of each and why heterogeneous backbones (representation diversity).

- 3.4.2 Dynamic attention-weighted fusion — the core novelty. Give the fusion equations: per-sample attention weights over the three [CLS]/pooled representations, softmax normalization, fused representation, classification head. Figure 2: architecture diagram of the fusion block.
- 3.4.3 Teacher training objective — cross-entropy, optimizer, any regularization.

3.5 Knowledge distillation to the student model

- 3.5.1 Student architecture — BERT-base + classification head; parameter count vs teacher.
- 3.5.2 Distillation objective — write the full loss: $L = \alpha \cdot \text{CE}(y, S(x)) + (1-\alpha) \cdot \tau^2 \cdot \text{KL}(\text{softmax}(T(x)/\tau) \parallel \text{softmax}(S(x)/\tau))$ (+ feature/hidden-state loss if used). Define α and τ and report the values you used.
- 3.5.3 Training protocol — 10 epochs, batch size, LR schedule, early stopping/checkpointing, mixed precision if used. Figure 3: KD schematic (teacher frozen $\rightarrow$ soft labels $\rightarrow$ student).

3.6 Evaluation protocol

- 3.6.1 Stratified 5-fold cross-validation — describe fold construction, per-fold training with resume/checkpoint support, and how fold metrics are aggregated (mean $\pm$ std). This turns your engineering (resume support on Kaggle) into a reproducibility statement.
- 3.6.2 Cross-dataset generalization — train on the primary dataset, evaluate zero-shot on Davidson; describe the label remapping needed to align class definitions (your "remapped dataset" notebook).
- 3.6.3 Explainability — SHAP setup: explainer type, background set, how token attributions are computed for transformer inputs; misclassification analysis procedure (sampling FP/FN, categorizing error types).

3.7 Computational complexity

Short subsection: parameters, FLOPs or inference latency of teacher vs student. This quantifies the deployment argument before the results section.

4. Experimental Setup (1–1.5 pages)

4.1 Datasets

Table 2: dataset statistics — name, source, #samples, class distribution, avg length, split sizes. Cover the primary training dataset and Davidson (cross-dataset). Cite dataset papers.

4.2 Baselines and compared methods

Include at least: (a) classical baselines (TF-IDF+LR/SVM) if available; (b) each single backbone fine-tuned alone (BERT, XLM-R, DeBERTa) — these are essential ablations; (c) simple ensemble baselines (majority vote / average); (d) the full teacher; (e) the KD student; (f) published SOTA numbers on the same dataset.

4.3 Evaluation metrics

Accuracy, macro/weighted Precision, Recall, F1, and (recommended) AUC. Justify macro-F1 for class imbalance. Define each briefly or cite.

4.4 Implementation details

Hardware (Kaggle GPU — state the GPU model, e.g., P100/T4), frameworks (PyTorch, HuggingFace Transformers versions), all hyperparameters in Table 3 (LR, batch size, epochs, τ , α , seed, max length, optimizer, scheduler). Reproducibility statement + code availability (GitHub link if you will release it — reviewers like this).

5. Results and Discussion (3–4 pages — the most important section)

Order results from internal validity → comparative superiority → external validity → interpretation. Every table gets 1–2 paragraphs of analysis; never drop a table without discussing it. The ESWA sample paper interleaves tables with explicit commentary on what each shows — do the same.

5.1 Main classification results (5-fold cross-validation)

Table 4: per-fold and mean±std results for teacher and student across all metrics. Figure 4: box/bar plot of fold-wise F1. Discuss stability (low std = robust).

5.2 Teacher vs. student: the distillation gain

Table 5: teacher vs student vs each single backbone. Headline: student (91.51%) > teacher (90.16%). **Explain why** — soft labels act as regularization / dark knowledge transfers inter-class similarity / born-again effect — and support with a citation and, ideally, evidence (e.g., smoother confidence distributions, lower calibration error, or reduced overfitting gap between train/val). Reviewers WILL ask for an explanation of student > teacher; pre-empt it here.

Add a statistical significance test (paired t-test or Wilcoxon over folds, or McNemar on predictions) for the student-vs-teacher difference. ESWA reviewers frequently request this; including it upfront avoids a revision round.

5.3 Comparison with state-of-the-art

Table 6: your methods vs published results on the same dataset (same splits where possible; footnote any protocol differences). 1–2 paragraphs positioning your result.

5.4 Cross-dataset generalization (Davidson)

Table 7: zero-shot performance on Davidson for teacher, student, and single backbones; include an in-domain Davidson reference number for context. Discuss the generalization drop honestly, what the label remapping required, and whether the student generalizes as well as the teacher (if yes — a second strong claim: distillation preserves generalization).

5.5 Ablation studies

- Fusion ablation: dynamic attention fusion vs average/concat/voting fusion.
- Backbone ablation: remove one encoder at a time from the teacher.
- KD ablation: student without KD (plain fine-tuning) vs with KD; effect of τ and α (small sensitivity table or line plot).

If some ablations were not run yet, run at least the "student without KD" one — it is the minimum needed to attribute the gain to distillation rather than to BERT-base itself.

5.6 Explainability and misclassification analysis (SHAP)

Figure 5: SHAP summary/force plots for representative correct predictions (which tokens drive "hate"). Figure 6: SHAP explanations for misclassified samples. Table 8: taxonomy of error types (sarcasm/irony, implicit hate, quoted hate, dialect/slang, annotation noise) with counts and examples (mask offensive tokens partially, e.g., f**k). Discuss what the errors imply for future work.

5.7 Efficiency analysis

Table 9: parameters, model size (MB), inference latency per sample (CPU and GPU), throughput — teacher vs student. This is the bridge into Section 6: the student is $\sim 3\times$ smaller and $\sim 3\times$ faster (your actual numbers), enabling real-time use.

5.8 Discussion

Synthesize: what the combined evidence (CV stability + student>teacher + cross-dataset + SHAP) says; practical implications for content moderation; threats to validity (dataset bias, English-only training, single-domain sources).

6. Real-Time Deployment: Streamlit Web Application (0.75–1 page)

Present deployment as a system contribution, not a demo. Structure:

- 6.1 System architecture — Figure 7: user input → (optional MarianMT Bangla→English translation for accessibility) → preprocessing → distilled student inference → prediction + confidence + SHAP token highlighting → UI. Describe each block in one sentence.
- 6.2 Interface and features — Figure 8: annotated screenshot(s) of the Streamlit UI showing input box, prediction, confidence, and explanation view. Mention the Bangla translation feature explicitly as an accessibility layer (input in Bangla is translated to English before classification), and note its limitation (translation errors propagate) — honesty here strengthens the paper.
- 6.3 Deployment performance — end-to-end latency per request, memory footprint, hosting details. One small table or inline numbers.

Framing rule: the deployed classifier is the ENGLISH monolingual student. The MarianMT step is an input-translation convenience, and you should state that Bangla performance depends on translation quality and is not a claimed contribution of the classification model.

7. Conclusion and Future Work (0.5 page)

First paragraph: restate problem, method, and the three strongest quantified findings (student > teacher with fewer parameters; robust 5-fold performance; cross-dataset behavior). Second paragraph: limitations (English-only training, binary/limited labels, translation-dependent Bangla support, dataset bias). Third paragraph: future work — native multilingual/bilingual training (this is where the "bilingual" ambition goes!), implicit-hate detection, larger student search, human-in-the-loop moderation. The ESWA sample paper ends with exactly this pattern: brief summary + explicitly stated future research directions.

8. Mandatory Elsevier/ESWA Back Matter

- **CRedit authorship contribution statement** — per-author roles (Conceptualization, Methodology, Software, Validation, Writing – original draft, Writing – review & editing, Supervision). Required; the sample paper includes it verbatim.
- **Declaration of competing interest** — standard sentence.
- **Data availability** — e.g., "The datasets used are publicly available at [links]; code is available at [GitHub]" .
- **Acknowledgments / Funding** — if applicable.
- **References** — ESWA uses APA-style (author–year) citations, ~50–80 references for a paper like this; heavy on 2021–2026 works.

9. Complete Figure & Table Plan (build these before writing)

#	Item	Content	Source notebook
Fig 1	Overall framework	End-to-end pipeline diagram	— (draw new)
Fig 2	Teacher architecture	DynamicFusionNet + attention fusion block	Monolingual_our_proposed_architecture
Fig 3	KD schematic	Teacher→soft labels→student, losses	bilingual-kd-10-epoch
Fig 4	CV stability	Fold-wise F1 box/bar plot	bilingual_5_fold_finally
Fig 5	SHAP correct	Token attributions, correct predictions	bilingual-kd-shap-misclassification
Fig 6	SHAP errors	Attributions for misclassified samples	bilingual-kd-shap-misclassification
Fig 7	Deployment architecture	Streamlit system diagram (with MarianMT)	Streamlit app
Fig 8	UI screenshot	Annotated app interface	Streamlit app
Tab 1	Gap analysis	Prior work vs features checklist	Related work
Tab 2	Dataset statistics	Sizes, classes, lengths	preprocessing cells
Tab 3	Hyperparameters	All training settings	training configs
Tab 4	5-fold results	Per-fold + mean±std, T & S	bilingual_5_fold_finally
Tab 5	Teacher vs student	vs single backbones + significance	kd-10-epoch
Tab 6	SOTA comparison	Published results, same dataset	literature
Tab 7	Cross-dataset	Zero-shot Davidson results	cross-dataset-remapped / monolingual-cross-data
Tab 8	Error taxonomy	Misclassification categories + examples	shap-misclassification
Tab 9	Efficiency	Params, size, latency, throughput	measure separately

10. Mapping Your Notebooks to Paper Sections

Notebook / asset	Goes into
Monolingual_our_proposed_architecture.ipynb	Sec 3.4 (teacher architecture), Fig 2, baseline rows of Tab 5
bilingual-kd-10-epoch-5-fold-cv.ipynb	Sec 3.5–3.6.1, Tab 4, Tab 5, Fig 3–4
bilingual_5_fold_finally.ipynb	Sec 3.6.1 & 5.1 (final CV protocol and numbers)
bilingual-kd-cross-dataset-10-remapped-dataset.ipynb	Sec 3.6.2 & 5.4, Tab 7 (label remapping description)
monolingual-cross-data-generalisation-first-attempt.ipynb	Sec 5.4 supporting analysis / ablation of remapping
bilingual-kd-shap-misclassification.ipynb	Sec 3.6.3 & 5.6, Fig 5–6, Tab 8
Streamlit app + MarianMT pipeline	Sec 6 entirely, Fig 7–8

11. ESWA-Specific Writing & Submission Tips

- Length: ESWA applied-ML papers typically run 12–20 double-column-equivalent pages, ~8,000–11,000 words, 6–9 figures, 8–10 tables. Your plan above fits this.
- Every claim needs a number or a citation. "Efficient" → give latency; "robust" → give std across folds.
- ESWA is an *applications* journal: reviewers reward the deployment section and real-world framing — do not shorten Section 6.
- Report mean $\pm$ standard deviation for all CV results and include a significance test for your central claim (student > teacher).
- Anticipated reviewer questions to pre-answer in the text: (1) Why does the student beat the teacher? (2) Is the gain significant or within fold noise? (3) Would plain BERT-base fine-tuning (no KD) reach the same score? (4) How were Davidson labels mapped and is the comparison fair? (5) Is the novelty of attention fusion sufficient vs existing fusion ensembles? — Sections 5.2, 5.5, and 2.5 exist to answer these.
- Use consistent naming throughout: "teacher (DynamicFusionNet)" and "student (distilled BERT-base)" — never switch to "bilingual model".
- Suggested writing order (fastest path): Sec 3 → Sec 4 → Sec 5 tables first, then prose → Sec 2 → Sec 1 → Abstract → Sec 6–7. Write the abstract LAST.
- Before submission: Highlights file, cover letter (1 page: problem, contribution C1–C5, why ESWA), suggested reviewers (3–5), and check ESWA's Guide for Authors for the latest template/word limits.